



Inspiring Excellence



CSE490 (D)

INTRODUCTION TO FOOTBALL DATA AND ANALYTICS

IN ASSOCIATION WITH BANGLADESH FOOTBALL FEDERATION

SPRING 2026



- Professional Certificate from BFF
- Internship Opportunity at **BFF Data Science** Department after successful completion
- Learning from the **industry experts** of Football Data Science
- Opportunity to **co-author** research papers for **Sports Data Conferences**

CSE490 (D)

Introduction to

Football Data & Analytics

Lecture-12

Off-Ball Metrics

Prepared by-
Tarvir Anjum Aditto
Lecturer,
CSE, Brac University



Off Ball Metrics

**“What do you do during 87 minutes when you do not have the ball?
That is what determines whether you’re a good player or not” - Johan Cruyff**



Building a Pitch Control Model (Recap)

Pitch control at a given location is the probability that a player (or team) will gain control of the ball if it moves directly to that location.

For a given location on a pitch:

1. How long would it take for the ball to arrive?
2. How long would it take for each player to arrive?
3. What is the total probability that each team will control the ball after it has arrived?

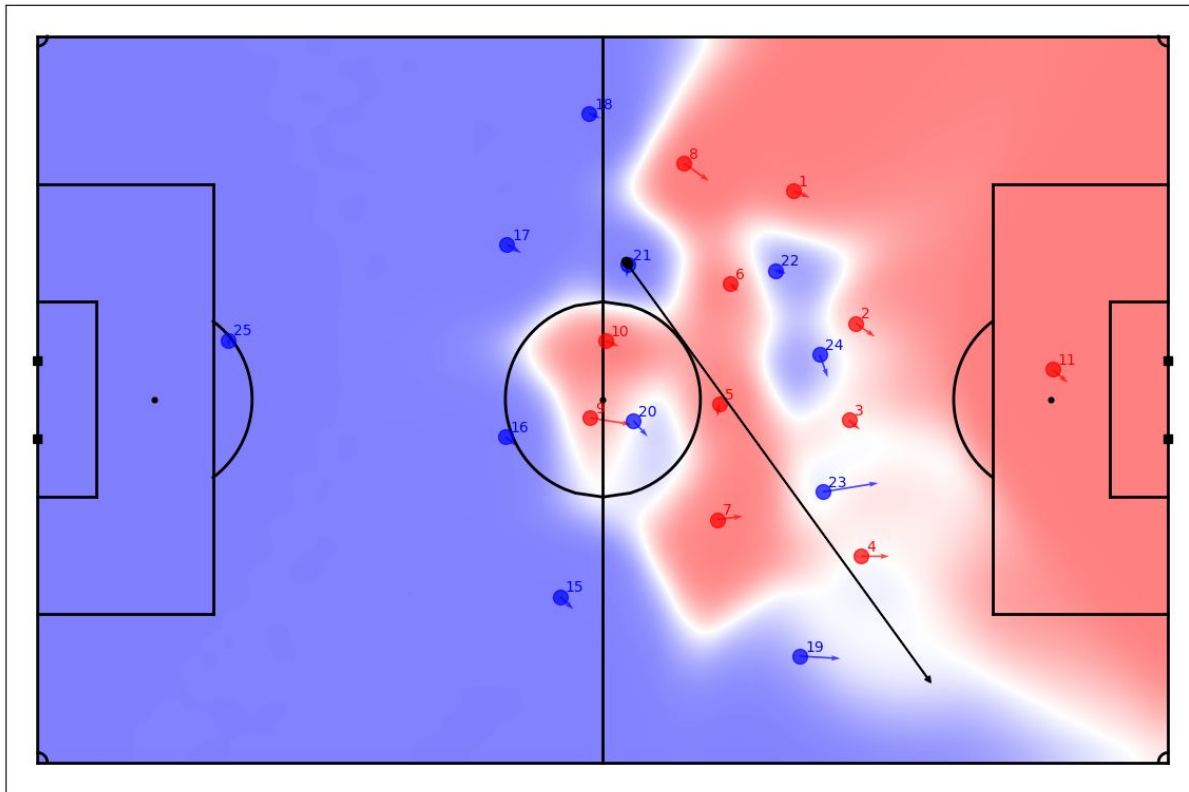
Repeat calculations for all locations on pitch

Based on Laurie shaw lecture– [Advanced football analytics: building and applying a pitch control model in python.](#)



Pitch Control

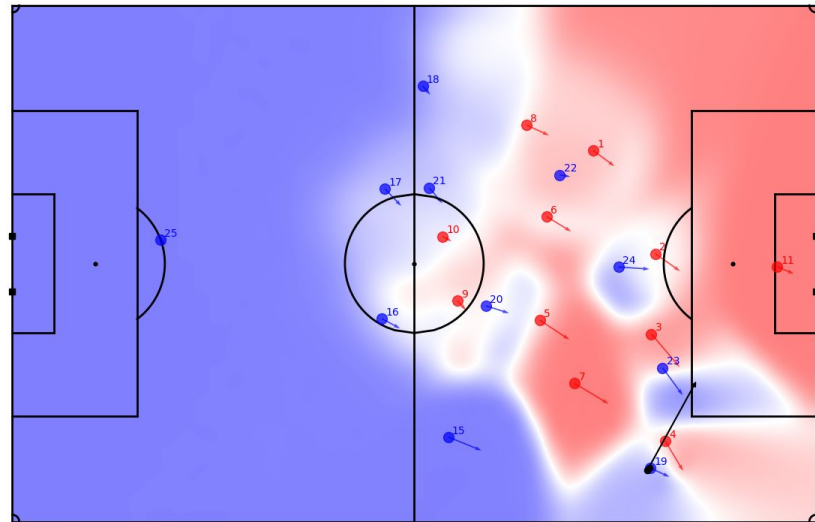
Pitch control — event 820



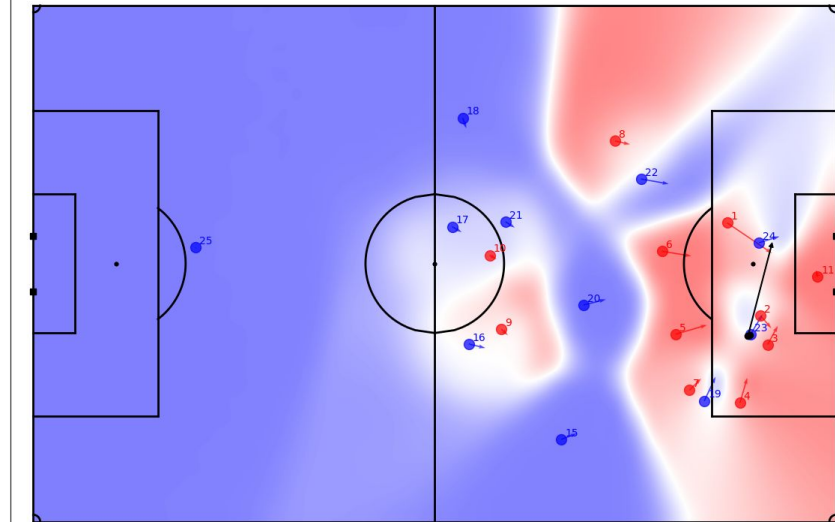
What do you think about this pass? Is it gonna be successful?

Pitch Control

Pitch control — event 821



Pitch control — event 822



After the successful pass, the forwards' run pulled defenders out of position, creating space for the attack.

Expected Possession Value (EPV)

The problem with pitch control only method is that possession at own half and possession at opponent box can have same amount of control, but we know possession at opponent box is more dangerous (more chance of scoring).

So, we combine **EPV** with pitch control.

EPV = P_poss(G | ball, team, opponents, match state)

Remember how we calculated position based xT (It is referred as EPV here).

–With the help of transition matrix (markov chain was one of the procedures)

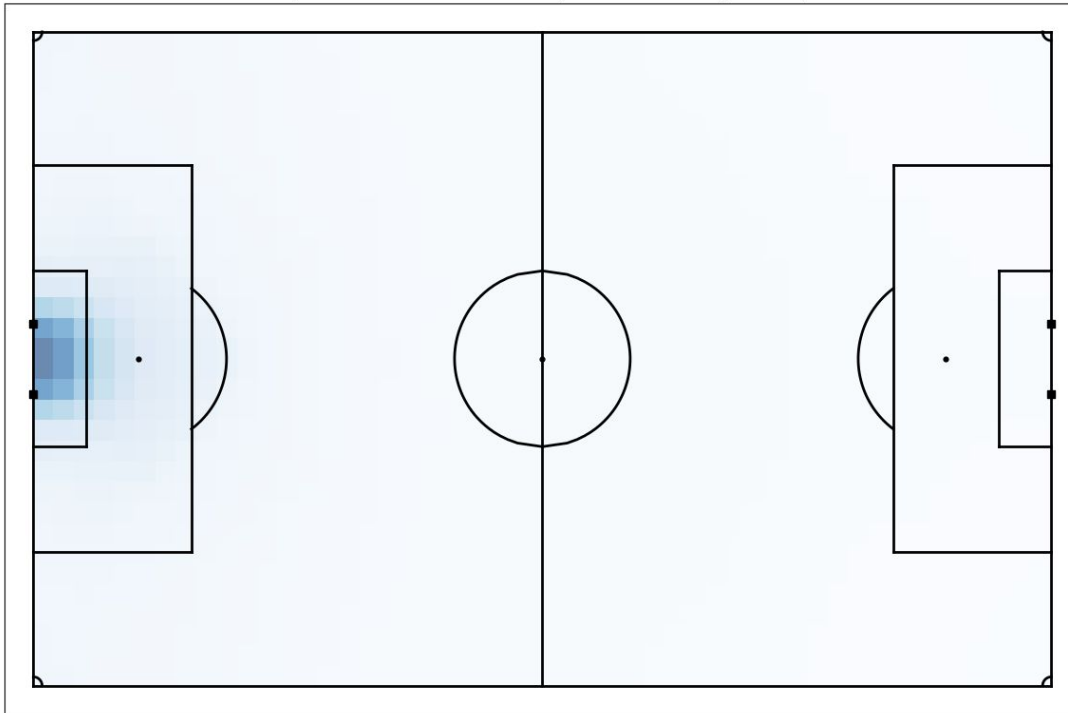
Value Added = Pitch Control(Rf) × EPV(Rf)

– Pitch Control(Ri) × EPV(Ri)



Expected Possession Value (EPV)

Expected Possession Value surface (home team attacking direction)

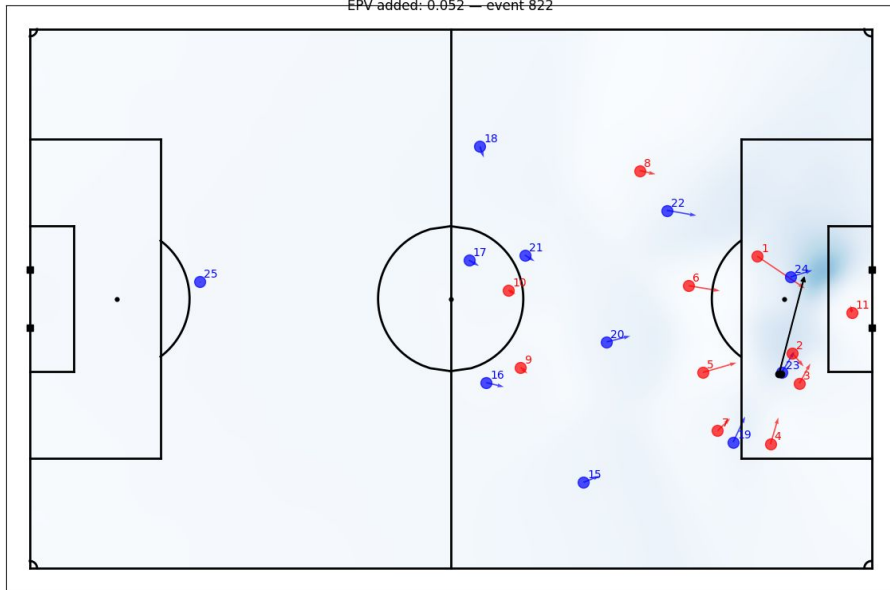


A team has more control close to their own goal, but less probability of scoring from there. On the other hand, the amount of control is low in front of opposition goal, but goal scoring probability is higher.

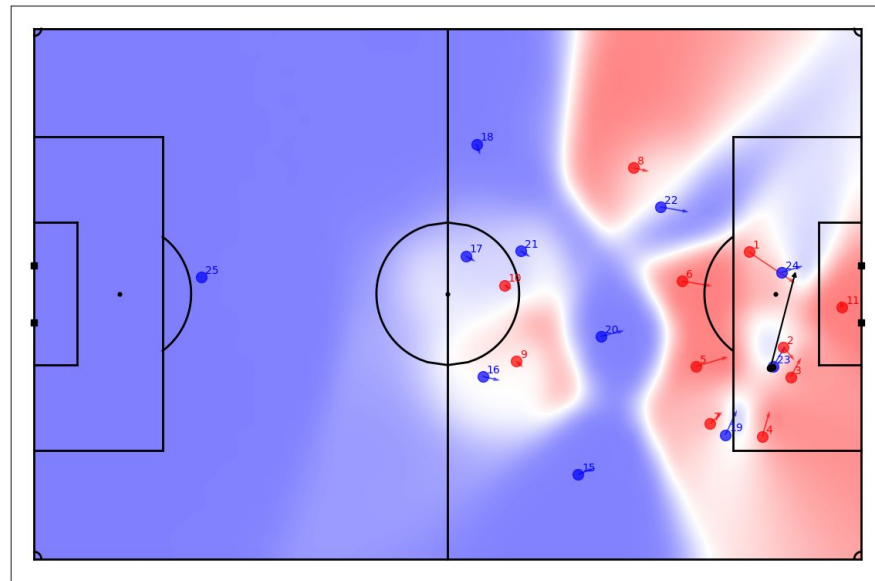
Expected Possession Value (EPV)

$$\text{EPV_added} = \text{EPV}(\text{after pass}) - \text{EPV}(\text{before pass})$$

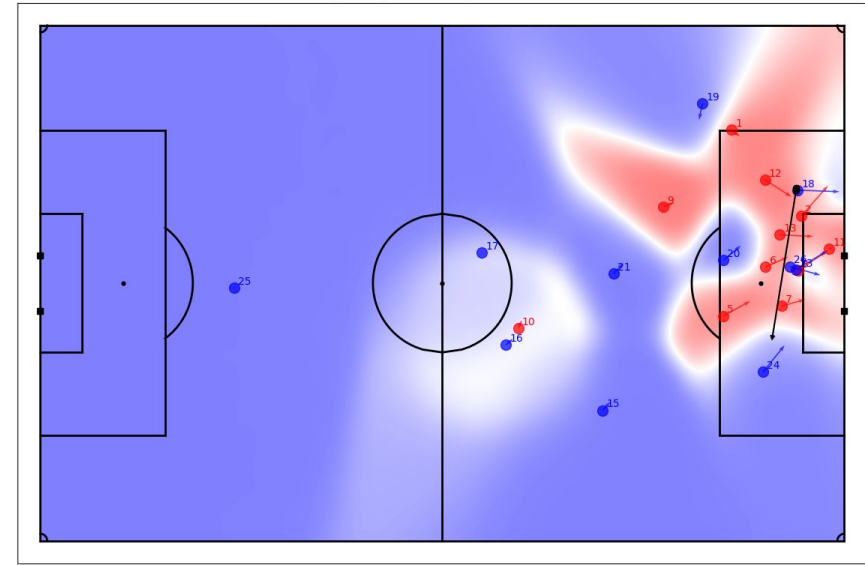
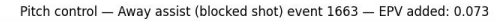
EPV added: 0.052 — event 822



Pitch control — event 822



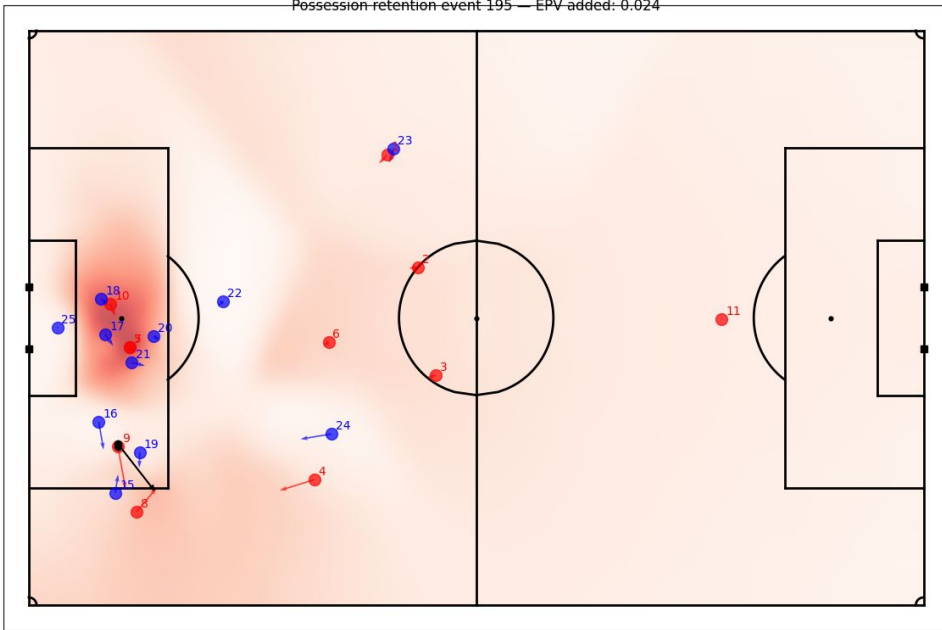
~~Away assist (blocked shot) event 1663 — EPV added: 0.073~~



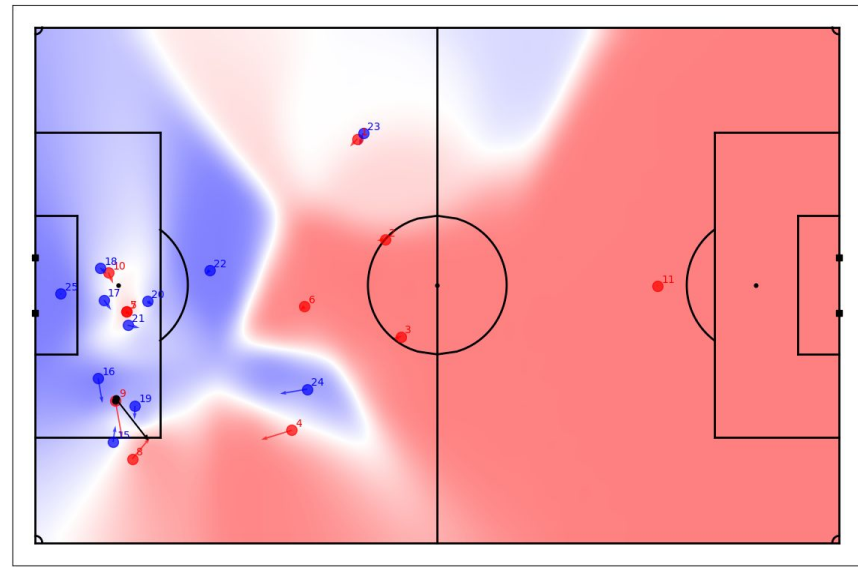
Expected Possession Value (EPV)

$$\text{EPV_added} = \text{EPV}(\text{after pass}) - \text{EPV}(\text{before pass})$$

Possession retention event 195 — EPV added: 0.024



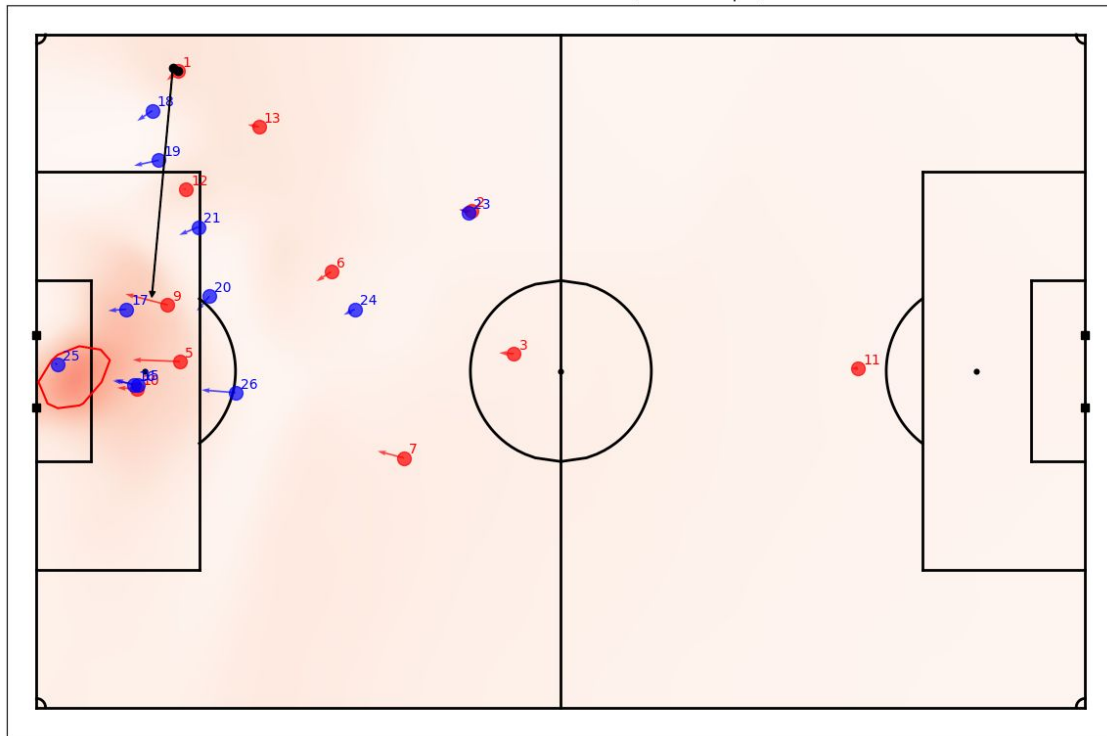
Pitch control — Possession retention event 195 — EPV added: 0.024



Best passing option

The best passing option is marked here as circle.

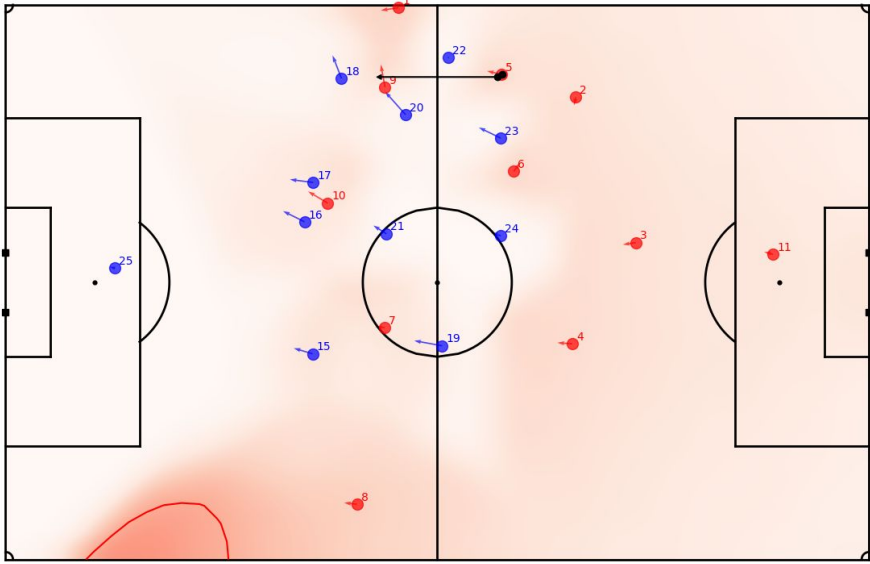
EPV with contours — event 1680 (assist example)



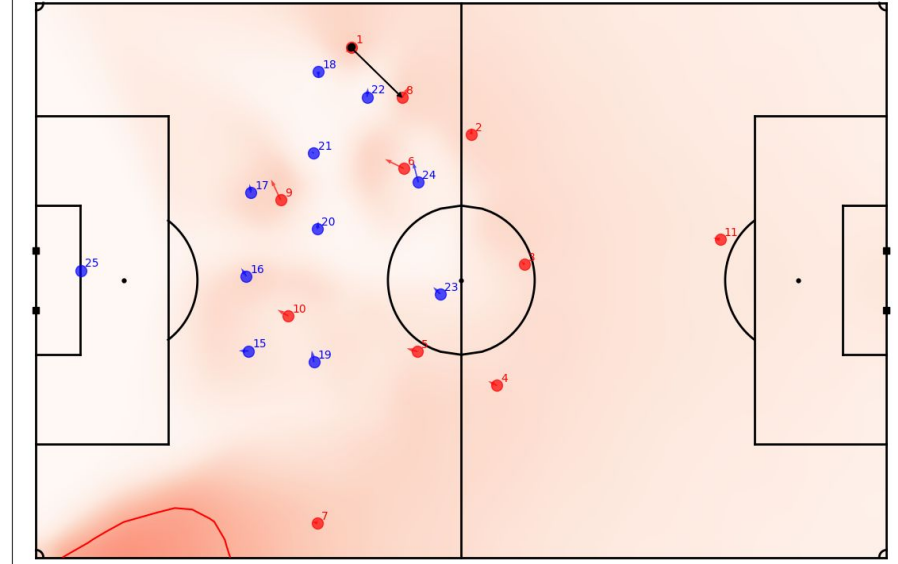
Cross Field Pass

Why cross field passes are appreciated much?

Cross-field pass — event 68



Cross-field pass — event 829



Off-Ball Runs

We can assign this value to the run, even if Rashford does not receive the pass or if (as was actually the case for this particular pass) if the pass is weighted poorly and Rashford has to slow down to receive it



Off-Ball Runs

Luke Shaw, who overlaps on the left wing. The two pictures show how he first accelerates and opens up a space on the left. In this case the pass doesn't come to him, but goes centrally. But the value comes both from the fact that the defenders have to move to close him down, opening up new spaces, and the fact that he provides an additional passing option

